Rules vs. Discretion in Monetary Policy

Based on Barro & Gordon (1983) and Romer Ch. 12

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Motivation: Two Stylised Facts

The rules vs. discretion debate was motivated by two empirical observations about monetary policy in the 1970s and early 1980s:

1. **Average rates of inflation and money growth were high** in most industrialised countries.
2. **Countries had a tendency to pursue activist, countercyclical monetary policies** — central banks routinely expanded the money supply (lowered interest rates) during recessions and contracted during booms.

Barro and Gordon (BG) suspected that fact (2) was causing fact (1): the habit of responding actively to the business cycle was itself generating persistently high inflation.

Key Concepts

Inflation (π_t) vs. the Interest Rate

These are *different* variables.

- **Inflation** π_t: the percentage change in the price level, $\pi_t = (P_t - P_{t-1})/P_{t-1}$.
- **Interest rate** i_t: the price of borrowing money. This is the central bank's *instrument* in reality.

The transmission mechanism in practice is:

$$\text{CB raises } i_t \;\Rightarrow\; \text{credit more expensive} \;\Rightarrow\; \text{less spending} \;\Rightarrow\; \text{lower demand} \;\Rightarrow\; \pi_t \text{ falls.}$$

In the BG model, this chain is **compressed**: the central bank is assumed to *directly choose* π_t. This is a simplification that allows the model to focus on the rules vs. discretion problem without worrying about intermediate channels.

Countercyclical Monetary Policy

The business cycle refers to fluctuations of output around its natural level $\bar{y}^n$:

- **Boom**: $y_t > \bar{y}^n$ — inflationary pressure.
- **Recession**: $y_t < \bar{y}^n$ — unemployment pressure.

A **countercyclical** policy acts *against* the cycle: expand (lower rates / print money) in recessions, contract in booms. The problem BG identified: this activist behaviour is anticipated by the public, generating an inflation bias even when there is no long-run output-inflation tradeoff.

Surprise Inflation

The central insight of rational-expectations models is that **only surprise inflation affects real output**.

If the public perfectly anticipates the central bank's actions, $\pi_t = \mathbb{E}_{t-1}(\pi_t)$, and there is no real effect. Only deviations $\pi_t - \mathbb{E}_{t-1}(\pi_t) \neq 0$ move output.

Model Overview

Assumptions

1. **Agents are rational**: they use all available information, including their knowledge of the central bank's incentives, when forming expectations.
2. **Agents anticipate the policy-maker**: since they know the CB's objective function, they can predict its optimal choice of π_t.
3. **The policy-maker directly controls inflation** π_t.
4. The natural level of output is **constant**: $y_t^n = \bar{y}^n$ for all t.

The Phillips Curve (Aggregate Supply)

Output deviates from its natural level only when inflation surprises agents:

$$y_t = \bar{y}^n + b(\pi_t - \mathbb{E}_{t-1}(\pi_t))\,, \qquad b > 0. \tag{1}$$

- y_t: actual output in period t.
- $\bar{y}^n$: natural (flexible-price) level of output — what the economy produces without frictions or monetary intervention.
- $b > 0$: sensitivity of output to inflation surprises.
- π_t: realised inflation.
- $\mathbb{E}_{t-1}(\pi_t)$: inflation expected by agents at $t-1$, *before* the CB acts.
- $\pi_t - \mathbb{E}_{t-1}(\pi_t)$: the **inflation surprise**.

Key implication: if $\pi_t = \mathbb{E}_{t-1}(\pi_t)$, then $y_t = \bar{y}^n$ regardless of the level of inflation chosen. The CB cannot raise output above $\bar{y}^n$ without surprising the public.

The Policy-Maker's Loss Function

The central bank minimises the following loss function each period:

$$Z_t = \frac{1}{2}(y_t - y^*)^2 + \frac{1}{2}\,a\,\pi_t^2, \tag{2}$$

where:

- y^*: the **output target** — the level of output the CB wishes to achieve. Crucially, it is assumed that $y^* > \bar{y}^n$, reflecting the fact that market imperfections (monopoly, taxes, frictions) make the natural level of output inefficiently low.
- 0: the **inflation target** — the CB wants zero inflation.
- $a > 0$: the **relative weight** on inflation stabilisation. A large a means the CB is more inflation-averse (conservative); a small a means it prioritises output.
- Both terms are *quadratic*: deviations above and below the target are equally costly, and large deviations are disproportionately costly.

Z_t is an abstract measure of social loss — not utility in the consumer sense. The CB seeks to make it as small as possible; $Z_t = 0$ only if both targets are hit simultaneously.

Substituting (1) into (2), the full problem is:

$$\min_{\pi_t} \; Z_t = \frac{1}{2}(\bar{y}^n + b(\pi_t - \mathbb{E}_{t-1}(\pi_t)) - y^*)^2 + \frac{1}{2}\, a\, \pi_t^2. \tag{3}$$

The Rule (Commitment) Solution

Setup

Under a credible rule, the CB *commits* in advance to a fixed path of inflation. Because the commitment is credible, agents anticipate the CB's choice perfectly:

$$\pi_t = \mathbb{E}_{t-1}(\pi_t).$$

The CB *internalises* this when minimising. Substituting into the Phillips curve (1):

$$y_t = \bar{y}^n + b(0) = \bar{y}^n.$$

Output is simply equal to the natural rate — **the CB cannot move output under a credible rule**.

Minimisation

The problem reduces to:

$$\min_{\pi_t} \; Z_t = \frac{1}{2}(\bar{y}^n - y^*)^2 + \frac{1}{2}\, a\, \pi_t^2.$$

The first term does not depend on π_t (it is a constant). The FOC is:

$$\frac{\partial Z_t}{\partial \pi_t} = a\, \pi_t = 0 \implies \pi_t^* = 0.$$

Result

$$\boxed{\pi_t^* = 0, \qquad y_t = \bar{y}^n.}$$

The value of the loss function is:

$$Z_t^* = \frac{1}{2}(\bar{y}^n - y^*)^2. \tag{4}$$

Interpretation: Z_t^* is the *unavoidable minimum loss*. The CB achieves its inflation target ($\pi = 0$), but cannot close the output gap ($\bar{y}^n - y^*$) — that gap exists because of real imperfections in the economy, not because of monetary policy. This is the **benchmark** against which discretion will be compared.

The Discretionary Solution

Setup

Under discretion, agents recognise that *no announced inflation path is credible* — the CB always has an incentive to deviate ex post. Therefore:

- The public does **not** believe any announcement.
- Inflation expectations $\mathbb{E}_{t-1}(\pi_t)$ are formed *before* the CB acts and are **taken as given** by the CB when it optimises.

The CB now solves (3) treating $\mathbb{E}_{t-1}(\pi_t)$ as a fixed number.

First-Order Condition

Differentiating (3) with respect to π_t:

$$(\bar{y}^n + b(\pi_t - \mathbb{E}_{t-1}(\pi_t)) - y^*)\, b + a\, \pi_t = 0.$$

Solving for $\hat{\pi}$:

$$\begin{aligned} b(\bar{y}^n - y^*) + b^2(\pi_t - \mathbb{E}_{t-1}(\pi_t)) + a\, \pi_t &= 0, \\ (a + b^2)\pi_t &= b(y^* - \bar{y}^n) + b^2\, \mathbb{E}_{t-1}(\pi_t), \\ \hat{\pi} &= \frac{b}{a + b^2}(y^* - \bar{y}^n) + \frac{b^2}{a + b^2}\, \mathbb{E}_{t-1}(\pi_t). \end{aligned} \tag{5}$$

Rational Expectations Equilibrium

Since agents are rational, they know the CB's optimisation problem and therefore know that the CB will choose $\hat{\pi}$ as in (5). In equilibrium:

$$\mathbb{E}_{t-1}(\pi_t) = \hat{\pi}.$$

Substituting back into (5):

$$\begin{aligned} \hat{\pi} &= \frac{b}{a + b^2}(y^* - \bar{y}^n) + \frac{b^2}{a + b^2}\hat{\pi}, \\ \hat{\pi}\left(1 - \frac{b^2}{a + b^2}\right) &= \frac{b}{a + b^2}(y^* - \bar{y}^n), \\ \hat{\pi} \cdot \frac{a}{a + b^2} &= \frac{b}{a + b^2}(y^* - \bar{y}^n). \end{aligned}$$

$$\boxed{\hat{\pi} = \frac{b}{a}(y^* - \bar{y}^n) > 0.} \tag{6}$$

Output Under Discretion

Since $\mathbb{E}_{t-1}(\pi_t) = \hat{\pi}$, the inflation surprise is zero:

$$y_t = \bar{y}^n + b(\hat{\pi} - \hat{\pi}) = \bar{y}^n.$$

Output is **still** equal to the natural rate — exactly as under the rule.

Value of the Loss Function

Substituting $\hat{\pi}$ and $y_t = \bar{y}^n$:

$$\hat{Z}_t = \frac{1}{2}(\bar{y}^n - y^*)^2 + \frac{1}{2}\,a\left(\frac{b}{a}(y^* - \bar{y}^n)\right)^2 = \frac{1}{2}(y^* - \bar{y}^n)^2 + \frac{b^2}{2a}(y^* - \bar{y}^n)^2.$$

$$\boxed{\hat{Z}_t = \frac{1 + b^2/a}{2}(y^* - \bar{y}^n)^2.} \tag{7}$$

Comparison: Rule vs. Discretion

	Inflation π	**Output y**	**Loss Z**
Rule	0	$\bar{y}^n$	$\frac{1}{2}(y^* - \bar{y}^n)^2$
Discretion	$\frac{b}{a}(y^* - \bar{y}^n) > 0$	$\bar{y}^n$	$\frac{1 + b^2/a}{2}(y^* - \bar{y}^n)^2$

Since $b^2/a > 0$ always:

$$\hat{Z}_t > Z_t^*.$$

The policy-maker is strictly worse off under discretion.

The Inflation Bias

The excess inflation under discretion:

$$\hat{\pi} - \pi^* = \frac{b}{a}(y^* - \bar{y}^n) - 0 = \frac{b}{a}(y^* - \bar{y}^n) > 0.$$

This is the **inflation bias** — strictly positive inflation with *no gain in output* relative to the rule.

- The CB tried to exploit the output-inflation tradeoff.
- The public anticipated this and raised $\mathbb{E}_{t-1}(\pi_t)$.
- In equilibrium, output is the same as under the rule, but inflation is higher.

What Determines the Bias?

$$\hat{\pi} = \frac{b}{a}(y^* - \bar{y}^n).$$

- **Larger** b (inflation is more effective at raising output) $\Rightarrow$ larger bias — the CB has more incentive to exploit the tradeoff.
- **Larger** $(y^* - \bar{y}^n)$ (bigger gap between target and natural output) $\Rightarrow$ larger bias — the CB is more desperate to close the gap.
- **Larger** a (more inflation-averse CB) $\Rightarrow$ smaller bias — the CB cares more about the inflation cost.

Summary

1. BG showed that **activist, countercyclical policy** under discretion generates an **inflation bias**: the CB inflates more than under a rule, but achieves the same output.

2. The source is **dynamic inconsistency**: the policy the CB would like to announce (low inflation) is not the policy it will actually implement once expectations are formed.

3. Under rational expectations, the public anticipates this, so the CB ends up with high inflation and no output gain — strictly worse than under a credible rule.

4. **Solutions** explored in the literature (Romer 12.7–12.8):

   - *Rules*: formal commitment mechanisms.
   - *Reputation*: repeated game — the CB builds credibility over time (Problem 4 of your problem set).
   - *Delegation*: appoint a more conservative central banker with weight $c > 1$ on inflation (Problem 3 of your problem set).